Project Title: AI Interview Preparation App

Overview

Getting ready for technical interviews can be stressful — but what if you had your own AI-powered mock interviewer available anytime?

This app helps users **practice coding interviews** by choosing topics (like **JavaScript**, **DBMS**, **Data Structures**, etc.) and selecting difficulty levels. Once set, the app uses **ChatGPT** to simulate a real Q&A interview session.

The best part? You can **review your answers**, **track your progress**, and **see detailed analytics** of your strengths and weak spots — all in a sleek React interface.

NOTE: ON THIS PLATFORM MAKE SURE NO ONE CAN MAKE ANY OTHER REQUEST OR PROMPT LIKE HOW TO MAKE TEA? ONLY IT SHOULD BE RELATABLE TO THE TECHNICAL INTERVIEWS

Tech Stack

Frontend:

- **React.js** – for building an interactive, responsive UI
- Use **React Router** for navigation and **Context API** or **Redux** for state management
- Charting libraries like **Chart.js** or **Recharts** to show analytics dashboards

Backend:

- **Node.js + Express.js** – to build REST APIs for questions, sessions, and user data
- **AI Integration**:
  → **OpenAI's ChatGPT API** (or similar) to generate interview questions and follow-ups

Authentication:

- **Passport.js** – with either **JWT** (for token-based auth) or **Session-based login** using cookies
- Protects user sessions, progress data, and review history

Database (Pick One):

- **MongoDB** – if you want flexible schema for storing AI responses and sessions
- **PostgreSQL** – if you prefer structured relationships and analytics across tables
- ORMs: **Mongoose** (MongoDB), **Prisma**/**Sequelize** (PostgreSQL)

Key Features

Topic & Difficulty Selection

- User chooses a topic (e.g., JavaScript, Algorithms, System Design)
- Then selects difficulty: **Easy**, **Medium**, or **Hard**
- Backend fetches relevant prompts and sends them to ChatGPT

AI-Powered Interview Simulation

- User enters a **mock interview session**
- Chat-like interface with **real-time question/answer flow**
- ChatGPT adapts based on user responses, like a real interviewer would

Store & Review Responses

- All sessions and user answers are stored

- Users can:
  → Review their past sessions
  → See correct or ideal answers
  → Compare their performance over time

Analytics Dashboard

- Visual insights into performance:
  → Topics with most correct/incorrect answers
  → Progress over time (weekly/monthly stats)
  → Success rate by difficulty level

- Built using charting libraries for a clean, visual experience

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Database Schema Overview

Collections/Tables:

- `users`
- `- id, name, email, password (hashed), preferences`
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- `interviews`
- `- id, user_id, date, topic, difficulty`
-
- `questions`
- `- id, interview_id, question_text, generated_by (AI), topic, difficulty`
-
- `answers`

- `- id, user_id, question_id, user_response, score (optional), feedback`
-
- `analytics`
- `- user_id, topic-wise summary, accuracy, improvement trends`

You can expand this to include:

- `feedback` – from AI or human mentor
- `bookmarks` – for questions the user wants to revisit later
